

RWTH Aachen University
Process and Data Science (PADS)

State Based Process Monitoring in Traditional Event Logs for Concept Drift Detection

Bachelor's Thesis

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Contents

1	Introduction	2
2	Related Work	2
3	Background	2
4	State Types	2
4.1	Intra-case State	2
4.2	Resource State	2
4.3	Inter-case State	2
5	Approach	2
5.1	Feature Extraction	2
5.2	Dimensionality Reduction	2
5.3	State Discovery	2
5.4	Drift Detection	2
5.5	Pipelines	2
6	Implementation	2
7	Evaluation	3
8	Conclusion	3

1 Introduction

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2 Related Work

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3 Background

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4 State Types

This thesis defines three complementary state types for traditional event logs, inspired by [1].

4.1 Intra-case State

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4.2 Resource State

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4.3 Inter-case State

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5 Approach

5.1 Feature Extraction

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5.2 Dimensionality Reduction

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5.3 State Discovery

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5.4 Drift Detection

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5.5 Pipelines

6 Implementation

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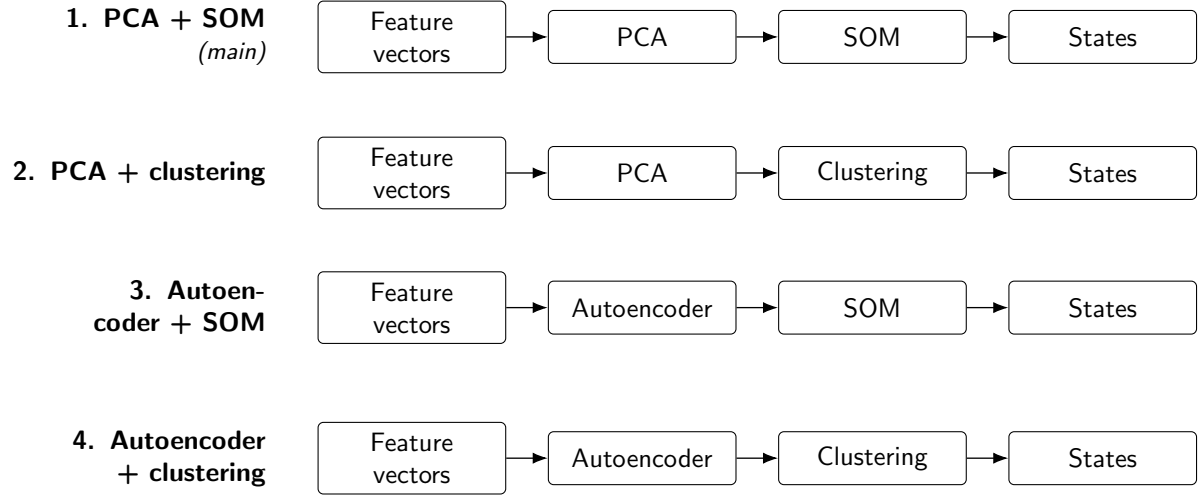


Figure 1: Pipelines for the four state-based concept drift detection approaches. Each approach combines a dimensionality reduction step (PCA or autoencoder) with a clustering step (SOM or k-means / similar) to produce discrete states from per-state-type feature vectors.

7 Evaluation

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8 Conclusion

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References

- [1] Alessandro Berti et al. “Identifying Execution States and Boundary Conditions in OCELS”. In: 2025.